# Bhabha scattering 3

The Casimir trick uses matrix arithmetic to sum over spin states.

From the first section we have

$$\sum_{abcd} \mathcal{M}_1 \mathcal{M}_1^* = \frac{e^4}{t^2} f_{11}$$
$$\sum_{abcd} \mathcal{M}_1 \mathcal{M}_2^* = \frac{e^4}{st} f_{12}$$
$$\sum_{abcd} \mathcal{M}_2 \mathcal{M}_2^* = \frac{e^4}{s^2} f_{22}$$

where

$$f_{11} = 8s^2 + 8u^2 - 64m^2 s - 64m^2 u + 192m^4$$
$$f_{12} = -8u^2 + 64m^2 u - 96m^4$$
$$f_{22} = 8t^2 + 8u^2 - 64m^2 t - 64m^2 u + 192m^4$$

By the Casimir trick we also have

$$f_{11} = \mathrm{Tr}\left((\not{p}_2 - m)\gamma^\mu(\not{p}_4 - m)\gamma^\nu\right) \mathrm{Tr}\left((\not{p}_3 + m)\gamma_\mu(\not{p}_1 + m)\gamma_\nu\right)$$
$$f_{12} = \mathrm{Tr}\left((\not{p}_2 - m)\gamma^\mu(\not{p}_4 + m)\gamma^\nu(\not{p}_3 + m)\gamma_\mu(\not{p}_1 - m)\gamma_\nu\right)$$
$$f_{22} = \mathrm{Tr}\left((\not{p}_3 - m)\gamma^\mu(\not{p}_4 + m)\gamma^\nu\right) \mathrm{Tr}\left((\not{p}_2 + m)\gamma_\mu(\not{p}_1 - m)\gamma_\nu\right)$$

Recall that slash matrices are formed as

$$\not{p} = p^\mu g_{\mu\nu} \gamma^\nu = p^0\gamma^0 - p^1\gamma^1 - p^2\gamma^2 - p^3\gamma^3$$

Eigenmath script